

Project: Azure Capacity Reservation Management Engine (ACRME)
Classification: Principal Cloud Architect - Architecture Governance
Version: 2.4
Date: 7 September 2026
Status: Accepted - supersedes ADR-001 v2.2 region-selection content; reconciled to Baseline v2.4
Part of: ACRME Architecture Decision Records - aligned to Capacity & Quota Management Requirements Baseline v2.4.

About ADRs. An Architecture Decision Record captures a significant architectural decision, the context that forced it, the options considered, the choice made, and its consequences. This ADR states the accepted region-selection decision matching the consolidated requirements baseline — notably the five-geography region model with per-geography distribution models and mandatory two-region CVAL/DR co-location (PLC-010a). Evidence tags: [Documented] , [Decided] , [Derived] , [Assumed] .

v2.4 reconciliation note — reservation-model gaps. Baseline v2.4 folds in the reservation-model gaps (seed-at-0 matrix CAP-022, per-AZ CRG structure CAP-023, reactive discovery CAP-024, Availability-Set ineligibility CAP-020/021, even zone-distribution PLC-011, naming convention OPS-006/C-12). **None of these change the region-selection or DR-placement decision recorded here** — the five-geography model, per-geography distribution models, exact-production-region-first placement, CustomerSeedRecord and two-region CVAL/DR co-location (PLC-010a) are all unchanged by the reservation-model work. The new per-AZ CRG structure (CAP-023) operates **with-in** a selected region and is specified in ADR-002/ADR-004, not here.

v2.4 (amended) note — Middle East DR now offered (cross-geo to Europe). As of the 11 Sep 2026 amendment to baseline v2.4, the Middle East DR_NOT_OFFERED position (former DEC-001) is **superseded**. Middle East DR **is now offered** as a **cross-geo DR** distribution model: **Prod and CVAL co-locate in a selected Middle East Standard region** (separate CRGs — co-location in a region is not capacity sharing, ENV-003) and **DR is placed cross-geo in a weighted-selected Europe Standard region**. Both the Middle East (Prod+CVAL) and Europe (DR) selections run through the **weighted capacity placement model**. New codes: **DR-020** (cross-geo DR ME→Europe), **PLC-010b** (cross-geo DR override to the PLC-010a co-location rule), **A-ME1** (data-residency assumption). This introduces a **third distribution model** — cross-geo DR — distinct from the US three-region model and the standard two-region co-located model. The CVAL-sacrifice DR bootstrap (DR-005/006) does **not** apply to the Middle East; ME DR uses dedicated reserved capacity in Europe. Europe now additionally serves as a **cross-geo DR destination** for the Middle East. The relevant sections below carry inline **[Amended v2.4]** markers.

ADR-001 - Region Selection and Customer Placement

Status: Accepted
Date: 27 August 2026
Deciders: Principal Cloud Architect, Platform Engineering, DR Owner, Product Platform
Related requirements: REG-001..REG-005, PLC-001..PLC-010, RDY-001..RDY-004, DR-014, DR-016..DR-019, DAT-002, DAT-005
Related constraints: ENV-003, CAP-011, CAP-012, QUA-012, NFR-002, NFR-010

Context

Requirements v2.1 changes the placement model from repeated geography-driven selection to a seeded, production-region-first model. Customers and product teams must know the exact Azure production region up front because a broad geography selection creates contract churn, data-residency ambiguity, and inconsistent product placement for the same customer. [Derived]

ACRME must still calculate CVAL and DR placement using current capacity, reservation, quota, zone, sharing, restriction, and DR-readiness state. The first valid decision becomes an authoritative seed that future products reuse. [Decided]

Key forces:

- Production protection is the primary objective. [Decided]
- Region, zone, quota, and capacity are hard isolation boundaries; capacity is never counted across regions or zones. [Documented]
- Region strategy is configuration-driven and spans **five supported geographies** (US, EU, Australia, Asia Pacific, Middle East), each catalogued with an explicit **distribution model** (baseline Section 6, REG-001). The US is the only three-region geography; EU, Australia and Asia Pacific are standard two-region (co-located CVAL/DR) geographies; **the Middle East is a cross-geo DR geography [Amended v2.4]**. New regions (e.g. Japan East, pending) are policy data added by config, not automatic rollout commitments. [Derived]
- **Middle East DR is now offered as a cross-geo DR model [Amended v2.4] (DR-020, PLC-010b; supersedes DR-014/DEC-001 DR_NOT_OFFERED)**. Per the 11 Sep 2026 amendment to baseline v2.4, **Prod and CVAL co-locate in a selected Middle East Standard region** (separate CRGs — ENV-003) and **DR is placed cross-geo in a weighted-selected Europe Standard region**. Both selections run the weighted capacity placement model. Data-residency for the DR copy is governed by assumption **A-ME1** (Europe as the approved cross-geo destination); per-country legal carve-outs remain possible but are out of scope of the automated engine. [Documented]
- Placement must return a fresh, machine-readable readiness state to AEP/provisioning and must fail safely on stale or incomplete state. [Decided]

Decision

Adopt a **production-region-first, seeded placement architecture**:

1. **Exact production region is the default input.** The default onboarding path requires the customer or platform workflow to provide the exact Azure production region. ACRME validates it rather than deriving it from a broad geography. [Decided]
2. **Geography-only selection is an exception path.** Geography input is retained only with explicit exception approval and customer acknowledgement that the selected production region becomes fixed until an approved migration changes the seed. [Decided]
3. **Customer seed record is authoritative.** The first placement decision writes `CustomerSeedRecord`; subsequent products/environments for the same customer and geography reuse it instead of re-running production selection. [Decided]
4. **CVAL and DR are selected after production is fixed.** Once the production region is validated, ACRME selects or validates CVAL and DR using current readiness, environment separation, restriction flags, workload distribution, quota/capacity state, and state freshness. [Derived]

5. **Placement creates readiness output, not just a region tuple.** Every evaluation returns a

`DeploymentReadinessResult` state:

text

READY | READY_WITH_RISK | QUOTA_DEFICIT | RESERVATION_DEFICIT |

CAPACITY_UNAVAILABLE | STALE_STATE | POLICY_BLOCKED | VALIDATION_REQUIRED

[Decided]

1. **Atomic holds prevent double-commit.** Committed placement creates a hold keyed by customer, region, zone, SKU/family, environment, policy version, and snapshot version. Conditional writes reject concurrent requests that attempt to consume the same headroom. [Decided]
2. **DR placement contributes to the DR index.** The seed drives the `SourceDestinationDRIndex` maintained by ADR-003 so a declared source-region failure activates only that source's mapped standby set. [Derived]

Region Classification and Policy

`PlacementPolicy` is the authoritative, versioned region catalogue. It includes region classification, supported geographies, **per-geography distribution model** (three-region, two-region co-located, or cross-geo DR — see below), zone support, SKU/family eligibility, separation class, approved cross-geo extension paths, `DR_NOT_OFFERED` flags, stale-state thresholds, weights, and exception metadata. [Decided]

Attribute	Requirement
Standard region	Eligible for automated placement and scoring.
Restricted region	Never recommended, scored, or auto-selected; usable only for explicit production-region exception.
Cross-geo DR destination	[Amended v2.4] A geography approved to host cross-geo DR for another geography's source regions. Europe is the approved cross-geo DR destination for the Middle East (A-ME1): the DR region is weighted-selected among Europe Standard regions (DR-020, PLC-010b), not a single fixed region.
DR_NOT_OFFERED	Produces seed value DR region = NOT_OFFERED ; ACRME must not silently substitute another geography. No longer set for the Middle East [Amended v2.4] — the former Middle East default (DR-014/DEC-001) is superseded by the cross-geo DR model (DR-020). The flag remains available for any future geography/country that Legal declares no-DR.
Policy version	Every change increments version and is recorded with approver, reason, effective date, and replay/audit metadata.

The number of regions per geography is set by the catalogue's **distribution model**, not a universal minimum. There are **three distribution models [Amended v2.4]**:

1. **Three-region (US).** Production, CVAL and DR each sit in a distinct in-geo region.
2. **Two-region co-located (EU, Australia, Asia Pacific).** Production is placed in one region and **CVAL and DR are co-located in the other region** (mandatory rule **PLC-010a**, via environment separation ENV-003). This delivers in-geo Prod + CVAL + DR **without** any cross-geo path — co-location is the standard model, not a fallback.
3. **Cross-geo DR (Middle East) [Amended v2.4]. Prod and CVAL co-locate in a selected Middle East Standard region** (separate CRGs; co-location in a region is not capacity sharing, ENV-003) and **DR is placed cross-geo in a weighted-selected Europe Standard region** (DR-020, PLC-010b). Both the source (Middle East Prod+CVAL) and destination (Europe DR) selections run the weighted capacity placement model. This model **overrides** the PLC-010a co-location rule for the Middle East: CVAL co-locates with **Prod locally**, not with DR, so the CVAL-sacrifice DR bootstrap (DR-005/006) does not apply — ME DR uses dedicated reserved capacity in Europe. The former Middle East DR_NOT_OFFERED position (DR-014/DEC-001) is superseded. [Derived]

Placement Flow

```

flowchart TD
    Request[Placement request] --> ExistingSeed{Seed exists?}
    ExistingSeed -- Yes --> Reuse[Reuse CustomerSeedRecord]
    ExistingSeed -- No --> Input{Input mode}
    Input -- Exact production region --> ValidateProd[Validate production region]
    Input -- Geography exception --> GeoApproval{Exception approved and acknowledged?}
    GeoApproval -- No --> PolicyBlocked[POLICY_BLOCKED]
    GeoApproval -- Yes --> DeriveProd[Derive production from Standard regions]
    ValidateProd --> ProdOk{Region valid, fresh, supported?}
    DeriveProd --> ProdOk
    ProdOk -- No --> NotReady[Readiness reason]
    ProdOk -- Yes --> SelectCVAL[Select or validate CVAL]
    SelectCVAL --> SelectDR[Select DR: in-geo, cross-geo Europe for Middle East, or NOT_OFFERED]
    SelectDR --> Fresh{State fresh?}
    Fresh -- No --> Stale[STALE_STATE]
    Fresh -- Yes --> Hold[Create atomic placement hold]
    Hold --> Seed[Write CustomerSeedRecord]
    Reuse --> Readiness[Return DeploymentReadinessResult]
    Seed --> Readiness
  
```

CustomerSeedRecord

`CustomerSeedRecord` must include:

- seed ID and customer/realm identifier;
- geography;
- production region;
- CVAL region;
- DR region or `NOT_OFFERED` ;
- products covered;
- region/zone/SKU-family policy context;
- decision timestamp;
- policy version and engine version;
- capacity/quota snapshot references and freshness;
- exception approval reference and customer acknowledgement, where applicable;
- active hold IDs;
- migration status and audit metadata. `[Decided]`

Seeds are not regenerated on upgrades, rebuilds, or routine deployments. Changes require an approved migration/exception workflow with impact analysis. `[Decided]`

Validation Rules

Rule	Check	Failure state
Exact Prod default	Request includes a specific production region unless an exception is approved.	POLICY_BLOCKED
Region catalogue	Region is present in active PlacementPolicy .	VALIDATION_REQUIRED
Standard automated path	Automated selection uses Standard regions only.	POLICY_BLOCKED
Restricted region exception	Restricted regions require explicit production-only request and approval.	POLICY_BLOCKED
DR_NOT_OFFERED	Geography/country flagged as no-DR writes NOT_OFFERED and blocks cross-geo substitution.	READY_WITH_RISK or POLICY_BLOCKED per policy
Region-model gate (geography-aware) [Amended v2.4]	Placement satisfies the geography's distribution model (REG-001): a three-region geography (US) requires Prod, CVAL and DR in three distinct regions; a two-region co-located geography (EU, Australia, Asia Pacific) requires Prod in one region and CVAL + DR co-located in the other (PLC-010a); a cross-geo DR geography (Middle East) requires Prod + CVAL co-located in a Middle East region and DR in a weighted-selected Europe region (DR-020, PLC-010b). A universal "three distinct regions" minimum is not enforced.	POLICY_BLOCKED
Freshness	Snapshot age <= configured maximum or synchronous refresh succeeds.	STALE_STATE
Capacity	Required reserved capacity exists or over-allocation is approved.	RESERVATION_DEFICIT or CAPACITY_UNAVAILABLE

Rule	Check	Failure state
Quota	Consumer/deploying subscription has required quota.	QUOTA_DEFICIT
Concurrency	Atomic hold commit succeeds.	VALIDATION_REQUIRED

Consequences

Positive

- Avoids production-region ambiguity and customer contract churn. [Derived]
- Produces stable placement across products for the same customer/geography. [Decided]
- Makes CVAL/DR decisions replayable from seed, policy version, and snapshot references. [Decided]
- Gives AEP/provisioning an explicit readiness contract instead of implicit success/failure. [Decided]

Negative / trade-offs

- Geography-only onboarding now needs exception governance and customer acknowledgement. [Decided]
- Seed migration becomes a governed workflow, not a simple re-run of scoring. [Derived]
- DR_NOT_OFFERED can create a readiness-with-risk outcome that must be handled commercially and operationally. [Derived]
- **[Amended v2.4]** Cross-geo DR for the Middle East adds a cross-geography dependency: Europe Standard regions now absorb Middle East DR load, and both the Middle East (Prod+CVAL) and Europe (DR) legs must be weighted-selected and held atomically. Data residency of the DR copy is governed by A-ME1 and must be re-validated if Legal changes the approved destination. [Derived]
- Atomic holds add state-management complexity but are required to prevent double-committed capacity/quota. [Decided]

Alternatives Considered

Alternative	Why rejected or constrained
Geography selection as default	Caused ambiguous customer intent and inconsistent product placement. [Derived]
Re-run placement per product	Risks drift across products for the same customer/geography. [Decided]
Use stale daily/weekly snapshots for deployment	Can deploy into capacity/quota that is no longer available. [Derived]
Auto-select cross-geo DR for two-region co-located geographies	Rejected. EU, Australia and Asia Pacific co-locate CVAL and DR in-geo (PLC-010a); cross-geo substitution is never silent for them. [Amended v2.4] Cross-geo DR is now the explicit, catalogued distribution model for the Middle East only (DR-020, PLC-010b): DR is weighted-selected among Europe Standard regions (not a single fixed region such as the former Switzerland North example), and the former DR_NOT_OFFERED seed is superseded. [Decided]
Fix a single Europe region (e.g. Switzerland North) for Middle East cross-geo DR	Rejected [Amended v2.4] . Baseline v2.4 requires the Europe DR region to be chosen through the weighted capacity placement model among Europe Standard regions so DR placement reflects live capacity/quota, not a hard-coded region. Switzerland North remains only a default example. [Decided]

Appendix - ADR Summary

ADR	Requirements Applied	Key Open Items
ADR-001 Region Selection and Customer Placement	REG-001..005, PLC-001..010a, PLC-010b , RDY-001..004, DR-014, DR-020, A-ME1	~~DEC-001~~ RESOLVED — Middle East DR now offered (cross-geo to Europe, v2.4 amended) ; DEC-003 geography exception approver; production stale-state threshold; per-country ME data-residency carve-outs (Legal)

Appendix - Status Legend

Status	Meaning
Proposed	Under discussion; not yet ratified
Accepted	Ratified and in force
Deprecated	No longer recommended but not yet replaced
Superseded	Replaced by a later ADR

Appendix - Evidence Tag Taxonomy

Tag	Meaning
[Documented]	Traceable to Azure platform behaviour or documentation
[Decided]	An explicit ACRME design choice recorded in this ADR set
[Derived]	A logical consequence of a documented constraint or decision
[Assumed]	Architectural judgement pending proof-of-concept validation

Related ADRs

- **ADR-002 - Quota and Capacity Management** (`ac-rme_adr_002_quota_and_capacity_management.md`)

- **ADR-003 - Capacity Management during Disaster Recovery (DR)** (`ac-rme_adr_003_capacity_management_during_dr.md`)
 - **ADR-004 - Forecast, Reconciliation, and Increase of Capacity and Quota** (`ac-rme_adr_004_forecast_and_increase_of_capacity_and_quota.md`)
-

Document Status: Accepted

Next Review: After DEC-003 decision, readiness API contract review, and first seed-record implementation test. (DEC-001 resolved — Middle East DR now offered as cross-geo DR to Europe, v2.4 amended 11 Sep 2026.)